INCH-POUND

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MIL-S-53046
15 August 1984

MILITARY SPECIFICATION

SURVEYING EQUIPMENT, ELECTRONIC DISTANCE MEASURING, MEDIUM-RANGE, (SEDME-MR)

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-S-53046. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers portable, medium-range, electro-optical distance measuring equipment (SEDME). Distances are measured using diode-emitted infrared modulated light waves. SEDME measuring unit will be solid-state construction and complete with accessories for unsupported operation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues f these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52298	- Tripod, Surveying; Extension-Leg-Type.
W-C-440	- Clip, Electrical, General Specification for.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	- Dyeing and Aftertreating Process as for Cotton Cloth.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-T-14132	- Theodilites, Surveying, Direction. Second Order.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

PUBLICATIONS

TB MED-524	- Occupational Environmental Health Control of Hazard to Health to from Laser Radiation.

(Application for copies should be addressed to the CMNDR, US ARMY PUBLCTN DISTRB CTR, 1655 WOODSON RD, ST LOUIS MO 63114.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description.

3.1.1 General. Survey Electronic Distance Measuring Equipment - Medium Range (SEDME-MR), hereinafter referred to as SEDME, shall consist of an electro-optical package, transport case or cases compatible retro-directive prisms, two 12-volt NiCad battery packs, charger, auxiliary external power cable not less than 20 feet in length, tribrachs, accessories and tools, and accessory case. When specified (see 6.2), a winterization kit, thermometer, altimeter-barometer, meteorological correction data, tripods, and backpack shall also be included. The SEDME shall measure distances to the accuracy specified in 3.12.2 by transmitting an amplitude-modulated light beam signal to, and receiving the signal back from, retro-directive prisms; then by a phase shift measurement of the transmitted and reflected modulated signal, or by "time of flight" measuring method (pulse modulated light), compute internally and display the corrected slope distance in meters. The SEDME shall have a built-in means for phase relationship comparison of the range frequencies or built-in means to measure the flight time of a light pulse. Distance measurement shall be done automatically under micro processor control. SEDME shall measure distances from 30 meters to 7 kilometers under the environmental conditions specified herein during day and night operation, and shall be mountable to the standard tripod specified in A-A-52298 by use of the universal tribrach specified in MIL-T-14132. The SEDME snall have an optical sighting telescope, automatic balance, audible aim, and tracking display.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.8).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7). Materials shall be free from all defects (see 4.5.1). and imperfections that may affect the operation or life of the instrument. Materials used shall be both lightweight and corrosion resistant, such as aluminum alloys, copper alloys, alloy steel containing a minimum of 11 percent chromium, or molded plastics. Where the use of corrosion-resistant materials is not practicable, the materials shall be treated to resist corrosion. Contact between dissimilar metals, as defined in MIL-STD-889, type II, that will cause deterioration of parts by galvanic corrosion shall be avoided, or dissimilar metals shall be insulated from each other as specified in MIL-STD-889. Parts and materials which are inherently resistant to fungus degradation shall be used whenever possible. Material of all bearing surfaces in contact with each other

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shall have the same coefficient of expansion or shall have provision for differential movement.

3.3.1 Material deterioration prevention and control. The SEDME shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the SEDME may be exposed.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the SEDME may be newly fabricated from recovered materials to the maximum extent practicable, provided the SEDME produced meet all other requirements of this specification Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the SEDME.

3.4 Environmental requirements,

3 4 1 Operational temperature. When in thermal equilibrium in air at any temperature from +130 to -5 °F, the SEDME shall operate with no external assistance, without damage, from an internal or external 12-volt power source, with the accuracy specified herein. The SEDME shall operate without damage and with the accuracy specified herein from 0 to -45 °F using an external winterization kit.

3.4.2 Storage temperature. The SEDME shall not be damaged by storage under the following conditions:

Air temperature - +160 °F - 4 hours.
Low temperature - -50 °F - 6 hours.

3.4.3 Solar radiation. When subjected to solar radiation of 355 Btu per square foot per hour, for not less than 4 hours in an environmental air temperature at 110 °F, +0 °F or -3 °F, the SEDME shall operate without damage within the accuracy specified herein.

3.4.4 Humidity. When subjected to relative humidities of at least 95 percent at air temperature varying between 68 ±2 °F and 120 ±2 °F, the SEDME shall not be damaged and shall operate within the accuracy specified herein.

3 4.5 Rain. The SEDME shall not be damaged nor show evidence of water penetration and shall operate within the accuracy specified herein before and after exposure to windblown rainfall of 3 ±1 in/hour at not less than 30 mph for a period of not less than 60 minutes.

3.4.6 Vibration. When subjected to the vibration test specified in 4.5.2.11, the SEDME shall not be damaged by vibration.

3.4.7 Shock. When subjected to the shock test specified in 4.5.2.10, the SEDME shall not be damaged.

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3.4.8 Magnetic environment. SEDME shall operate and measure distance within the accuracy specified in 3.12.2 and time specified in 3.12.3 while in the presence of a magnetic field of 3 gauss. The magnetic filed orientation shall be east-west, north-south, and vertical.

3.4.9 Sand and dust. When the SEDME and transport case are subjected to blowing sand and dust, the SEDME shall show no evidence of sand and dust penetration into the optics or electronics package, or any lens abrasion. The transport case shall show no evidence of sand or dust penetration.

3.4.10 Atmospheric conditions. The SEDME shall measure slope distances of 7 km in excellent atmospheric conditions (overcast, no haze, visibility about 30 km, no heat shimmer).

3.5 Reliability. When tested as specified in 4.5.2.15, for the system and 300 hours for the continuous electronic cycling, the specified Mean-Time-Between-Failure (MTBF) of the SEDME shall be 20 missions.

3.6 Safety. SEDME shall present no uncontrolled hazards to operators, bystanders, or maintenance personnel, and shall conform to all applicable requirements of TB MED 524 and requirement 1 of MIL-STD-454. SEDME shall meet either the power and irradiance requirement of TB MED 524 for a class I device, or be less than the irradiance criteria for a class I device and be less than five times the power limitations for a class I device (see 6.7 for safety ordering data).

3.7 Human factors. SEDME shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, and test equipment (see 6.3 and 6.4) shall be adequate to perform the intended function of assisting or conducting maintenance operations during testing as specified in section 4.

3.8.2 Maintenance ratio. When tested as specified in 4.5.2.16.2, the SEDME shall have a maintenance ratio of not more than 0.05. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8.3 Ease of maintenance. SEDME construction shall provide ease of access to batteries and fuses (or reset buttons) such that tools are not required to connect a battery or external power nor to change a fuse. SEDME construction shall provide a self-test display to verify satisfactory operating capability prior to initial distance measurement.

3.9 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the SEDME shall conform to MIL-STD-461, methods RE02 for radiated and CE03 for conducted tests, for class B equipment. The frequency spectrum scan shall be from 0.014 through 1 GHz during the radiated tests and from 0.020 through 50 MHz during the conducted tests with the following exceptions: (a) Broadband radiated emissions (RE02) at 0.3 MHz and 0.7 MHz shall

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be increased 5 db and, (b) Broadband radiated emissions (RE02) at 150 MHz shall be increased 6 db above the requirements of MIL-STD-461.

3.10 Treatment and painting. The portions of SEDME normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the type and color shall be gree 383.

3.10.1 Chemical resistant paint. When specified (see 6.2) the chemical resistant paint shall conform to MIL-T-704, type G

3 11 Physical characteristics

3.11.1 Portability. SEDME shall be furnished in man-portable packages.

3.11.2 Weight and size. The transit weight (exclusive of tripods but including primary power source) shall be not more than 25 pounds and the transport weight (exclusive of tripods but including primary power source and transport case) shall be not more than 75 pounds.

3.11.3 Controls. Panel dials, knobs and switches shall be functionally grouped and constructed for simple operation. The front panel of the SEDME shall consist of not less than the following functions:

a. Off-on.
b. Battery strength.
c. Self-check.
d. External signal return.
e. Measurement actuator.
f. Slope distance display.
g. Illumination for control panel.
h. Tracking mode.

3.11.4 Power SEDME shall operate from an internal 12-volt dc NiCad battery or from an external 12-volt dc battery conforming to MIL-B-11188, class FE, lead-acid via the external power cable. SEDME shall contain built-in protection against reverse polarity from the power source. SEDME winterization kit heating elements shall operate from batteries conforming to MIL-B-11188, 12-volt dc, class FE.

3.11.4.1 Main power source. SEDME shall be provided with a nickel-cadmium battery or battery pack with a combination of cells to give 12-volt operation. The battery shall be an internal or external power source. The battery shall be rechargeable via a 110/120 volt, 50/60 Hz charger supplied with the system and shall provide at least one hour of measuring time at 68 °F (20 °C). The charger will include an adapter for connection to 220 volt power supplies in Europe. When tested as specified in 3.4.7, the battery(s) shall be spillproof; shall have no detectable liquid or gas regardless of operation or storage orientation; shall not be damaged. The battery shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

3.11.4.2 External power sources. SEDME shall operate off lead-acid, storage batteries conforming to MIL-B-11188, 12-volt dc, class FE for 4 hours, and off 1' volt dc batteries mounted in a military vehicle.

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3.11.4.3 External power cable. The external power cable shall be not less than 20 feet long and shall terminate in battery clips conforming to W-C-440, type PC, with insulators.

3.11.4.4 Battery charger. A battery charger capable of charging at least two 12-volt 2-amphour batteries, or a built-in battery charger with each NiCad battery, shall be furnished.

3.11.5 Sighting telescope. The SEDME sighting telescope shall have not less than 20 diameters magnification power and a crosswire reticle for sighting purposes. The reticle shall be opaque in appearance, and be parallel to the transmitting optics. The telescope eyepiece shall focus uniformly and without back-lash or binding. The covers for the lenses shall conform to ASTM D 2000, class M3BA514A14B13C12F19Z. A means of colliminating the telescope line of sight with the electronic line of sight shall be provided.

3.11.6 Horizontal and vertical axes. The horizontal and vertical axes shall be suspended or supported on bushings or bearings. Bearings shall be of the free-movement type with a motion clamp and a tangent screw for slow motion control and precise adjustment. Movement through the vertical tilt limits, minimum ±45 degrees, and the horizontal plane shall not show signs of backlash, binding, and endplay. All bearings, motion clamps and screws shall be protected against moisture, dust, and deleterious foreign matter; the bearings, motion clamps and screws need not be sealed.

3.11.7 Electronics package.

3.11.7.1 Modules. The SEDME electronics package shall be constructed in modular units and shall have a positive guide arrangement to facilitate removal and replacement of the module.

3.11.7.2 Panel marking. Markings on the control panel shall identify each control, meter, and connector by its functional name, symbol, or operating instruction reference.

3.11.7.3 Illumination. The SEDME instrument panel shall have internally mounted illumination for night operation (see 3.11.3). All external lighting shall be blue-green color between 400-700 nanometer wavelengths with no light visibility beyond a 50-meter circle with the instrument as center point under moonless and starless night conditions. All external visible light produced shall be intensity adjustable.

3.11.7.4 Displays. Light emitting diode or liquid crystal technology shall be used for numerical displays.

3.12 Performance.

3.12.1 Precalibration. Each SEDME shall be calibrated by the contractor so that there shall be no instrument constants or retrodirective prism offsets that cannot be corrected in the instrument prior to distance measurement. If such constants or offsets do exist, they shall be stamped on the applicable hardware in a conspicuous place to facilitate presetting SEDME. A means shall be provided for adjusting SEDME to the proper constant without use of special tools.

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3.12.2 Accuracy and range.

3.12.2.1 Excellent atmospheric conditions. The SEDME shall measure distances from 30 m to 7 km in excellent atmospheric conditions (see 3.4.10) with a standard error not to exceed ±(5 mm + 5 mm/km) (after entering meteorological data or correction factor) when used in a system of one electro-optical package (distance meter) and one prism assembly.

3.12.3 Distance measurement time. After setup and actuation of the SEDME off-on power switch to the "ON" position, the time required for an initial distance reading in ambient temperatures from 120 to -5 °F shall be not more than 30 seconds. Subsequent readings shall be accomplished in not more than 6 seconds. The same time limitation shall apply to SEDME after having been warmed from -45 to -5 °F by the winterization kit (see 3.17).

3.12.4 Warmup time. SEDME shall operate within the range, time, and accuracy specified herein after warmup by the winterization kit, when applicable. The winterization kit (see 3.17) shall be capable of heating SEDME to operating temperature from -25 °F within 25 minutes and from -45 °F within 45 minutes. An external power source (see 3.11.4.2) can be used.

3.12.5 Stability. SEDME shall not drift beyond the accuracy specified herein over the battery operating life. SEDME shall have a battery strength indicator (see 3.11.3) which shall show 12 volt dc power available, or battery light. This indicator shall either monitor continuously or be a part of the prescribed measuring sequence.

3.12.6 Beam interruption. SEDME shall operate within accuracy and range specified herein even though the transmitted light beam is interrupted by objects moving randomly into and out of the optical path. Under this condition, the measuring time may exceed that specified in 3.12.3. Under no condition shall SEDME, when operated in accordance with the operator's manual, derive a distance measurement when the light path is permanently blocked by an object or when there is no return light from a reflector.

3.13 Reflector. Two retro-reflecting sets shall be furnished as a part of the SEDME. Each set shall consist of:

a. A triholder and three prisms.
b. A holder and necessary prisms to achieve the required 7 km range.
c. Two adapters for mounting the prism holders in the universal tribrach specified in MIL-T-14132. The prism adapter shall permit the prisms to be rotated through 360 degrees about the vertical axis without disturbing either the tribrach or plumb of the assembly over the survey mark. A clamping device or friction fit shall also be provided for locking the prism at any point after rotation.

3.14 Tripod. When specified (see 6.2), three tripods conforming to A-A-52298 shall be furnished.

3.15 Backpack. When specified (see 6.2), a means for transporting SEDME by backpack shall be provided. The means for backpacking shall be human engineered for ease and comfort of a small- to large-framed, 145-pound to 225-pound individual. Canvas, if used, shall conform to CCC-C-419, type and weight to suit

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the purpose, and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f, commercial color. The standard military theodolite backpack specified in MIL-T-52114 is acceptable as a SEDME backpack.

3.16 Cases.

3.16.1 Transit-transport case. The transport case (housing the electro-optical package, accessories and tools) shall be of metal or reinforced plastic laminate, or molded plastic designed with reinforced ribs, and shall have a removable inner liner which is molded to accept the various SEDME accessories. The case shall withstand operating, transport, and storage usage and shall be adequately blocked and padded to prevent damage that might otherwise result if the transport case is dropped from a distance of 36 inches, and to protect the instrument against damage, during field transport, shipment, and storage, from rainfall, shock, and vibration as specified herein. Each case shall contain a desiccant. The case shall open in such a way that hardware will not fall out when the case is opened. The case shall be provided with heavy-duty locking catches and, where possible, shall provide means for padlocking. Fixtures shall be so installed inside the case as to prevent movement and as to adequately protect spare parts and accessories during storage and transport. All items listed in 3.18 (except the backpack) shall be stored in the transit-transport case.

3.16.2 Accessory case. An accessory case or cases shall be provided for each retro-reflecting set (see 3.13). The case shall conform to the same requirements as the transit-transport case.

3.17 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished for use with SEDME at temperatures below -5 °F (see 3.12.4).

3.18 Accessories and tools. The following shall be the minimum furnished with each SEDME as accessories and tools (see 3.16.1):

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Item	Quantity
Spare fuses	4
Meteorological correction chart (metric)	2
Brush, lens, camel's hair	1
Tissue, lens	1 pkg
Screwdriver, flat tip, jeweler's	1
Pins, adjusting, alinement	1
Thermometer, centigrade (contractor's commercial standard) (see 6.2)	1
Altimeter-barometer, mm of Hg graduation (supplier's commercial standard) (see 6.2)	1
External power cable (contractor's commercial standard-minimum length 20 feet)	1
Backpack (contractor's commercial standard) or theodolite backpack (see 3.15 and 6.2)	1
Instruction manual(s)	1 ea
Covers, telescope (both ends) lens	1 set
Cover, instrument (optics) lens	1 ea
Battery (NiCad), or packs	2 ea
Charger, (NiCad); if not built into battery pack	1 ea

3.19 Identification marking. The SEDME shall be identified in accordance with MIL-STD-130. The marking shall be applied to the SEDME on labels conforming to MIL-P-514, type I, style 3, composition C, of type I, grade A, class 1 material. Plates shall be attached by adhesive binder in a location where the plate will be both visible and legible.

3.20 Interchangeability. All parts of the same manufacture having the same system part number or function shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.21 Workmanship. All parts, components, and assemblies of the SEDME instrument, tripod, and cases including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful extraneous material. All parts shall be finished. There shall be no cracks, sharp bends, dents, or chips in finished parts. There shall be no loose solder or insecure electrical connections. The fit of assemblies and subassemblies shall be made without imposing strain or forcing parts. All threads requiring movement and all sliding and rotatable parts shall move without binding, backlash, or endplay. Clamping actions, when required, shall be positive and when unclamped, shall not cause drag. No lateral motion shall exist between the electro-optics housing and support. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between glass and coated film or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Bearing surfaces shall show no scratches, or any irregular surface blemishes. External surfaces shall be free from burrs and sharp edges, and corners shall be chamfered and rounded except when sharp edges and corners are required.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article SEDME shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article SEDME shall be subject to the tests marked "X" in table I, column 1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be subject to the tests marked "X" in columns 2 of table I. Presence of one or more defects shall be cause for rejection.

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4.5 Inspection procedure.

4.5.1 Examination. The SEDME and associated equipment shall be examined as specified herein, and immediately after each test, for the following defects:

101. Parts, components, and accessories missing (see 3.1.1)
102 Parts, components, and accessories not as specified (see 3.1.1).
103. Beam interruption not as specified (see 3.1.1).
104. Materials not as specified (see 3.3).
105. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the SEDME (see 3.3.2).
107. Safety not as specified (see 3.6).
108. Hazardous infrared radiation present (see 3.6).
109. Manuals not as specified (when required) (see 3.8.1).
110. Accessibility of parts and components for service and maintenance not as specified (see 3.8.3).
111. Test points not provided for electrical circuitry.
112. Treatment and painting not as specified (see 3.10).
113. Dimensions and weight not as specified (see 3.11.2).
114. Controls not as specified (see 3.11.3).
115. Power source not as specified (see 3.11.4).
116. Battery charger not as specified (see 3.11.4.4).
117. Sighting telescope not as specified (see 3.11.5).
118. Horizontal and vertical axes not as specified (see 3.11.6).
119. Leveling screws not as specified (see 3.11.6).
120. Clamps and tangents not as specified (see 3.11.6).
121. Instrument modules not as specified (see 3.11.7.2)
122. Illumination not as specified (see 3.11.7.3).
123. Prism assemblies not as specified (see 3.14).
124. Tripods not as specified (see 3.14).
125. Backpack portability not as specified (see 3.15).
126. Cases not as specified (see 3.16).
127. Winterization kit not as specified (see 3.17).
128. Accessories and tools not as specified (see 3.18).
129. Identification markings not as specified (see 3.19).
130. Interchangeability not as specified (see 3.20).
131. Workmanship not as specified (see 3.21).

4.5.2 Test schedule. The test schedule shall be in accordance with table I.

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TABLE I. Test schedule.

First article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	X	Examination	4.5.1	3.1
X	X	Accuracy	4.5.2.2	3.12
X	-	High temperature operation	4.5.2.7.2	3 4.1
X	-	High temperature storage	4.5.2.7.3	3.4.2
X	-	Performance	4.5.2.3	3.12
X	-	Solar radiation	4.5.2.7.4	3.4.3
X	-	Stability	4.5.2.4	3.12
X	-	Humidity	4.5.2.8	3.4.4
X	-	Performance	4.5.2.3	3.12
X	-	Low temperature operation	4.5.2.7.5	3.4.1
X	-	Low temperature storage	4.5.2.7.6	3.4.2
X	-	Performance	4.5.2.3	3.12
X	-	Rainfall	4.5.2.9	3.4.5
X	-	Performance	4.5.2.3	3.12
X	-	Shock	4.5.2.10	3.4.7
X	-	Performance	4.5.2.3	3.12
X	-	Vibration	4.5.2.11	3.4.6
X	-	Performance	4.5.2.3	3.12
X	-	Magnetic environment	4.5.2.12	3.4.8
X	-	Performance	4.5.2.3	3.12
X	-	EMI	4.5.2.13	3.9
X	-	Performance	4.5.2.3	3.12
X	-	Sand and dust	4.5.2.17	3.4.9
X	-	Performance	4.5.2.3	3.12
X	-	Power consumption	4.5.2.5	3.11.4
X	-	Battery life	4.5.2.6	3.11.4.1
X	-	Human engineering	4.5.2.14	3.7
X	-	Reliabiltiy	4.5.2.15	3.5

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TABLE I. Test schedule. (continued)

First article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	-	Maintainability	4.5.2.16	3.8
X	-	Accuracy	4.5.2.2 (1-2-3 only)	3.12
X	-	Secure lighting	4.5.2.18	3.11.7.3

4.5.2.1 Condition of test. Prior to beginning the test, the contractor shall prepare SEDME to operate as specified herein.

4.5.2.2 Accuracy, range, and time. SEDME conformance to 3.12.2 (accuracy and range) and 3.12.3 (distance measuring time) shall be tested as per sequence in table I. It is permissible to reject one reading out of the group of 10 required for a distance measurement. More than one reject in a distance measurement shall be cause for rejecting all 10 readings and initiating a new measurement. Rejection of two consecutive distance measurements shall in any case (either reference line or baseline), constitute failure of the test. No allowance shall be made for plumbing errors in test set-ups. The accept/reject criteria herein has already provided for the maximum allowable error. It is permissible to reduce the number of retro-reflecting prisms at distances less than 200 meters in order to eliminate possible off-set errors and prevent "saturating" the instrument receiver. The SEDME shall be tested for accuracy, range, and time in the following set-ups:

a. A baseline consisting of a master monument and a 10-meter beam, graduated at each meter, respectively, with one of the meter spacings further graduated in 10 equal spacings, shall be used. The graduated beam shall be placed with the zero end at a distance of 30 meters from the master monument, then aligned radially away from the master monument. Environmental conditions for this test shall be stable. SEDME shall be positioned over the master monument at a height coincident (± 0.10 meter) with the retrodirective prism, mounted on the beam. Distance measurements shall then be made at 0 through 10 meters in 1-meter increments and corresponding time noted. Accuracy and corresponding time measurements not conforming to 3.12.2 and 3.12.3 shall constitute failure of the test.
b. A baseline consisting of a master monument and eight monument stations set on line 100 to 150 meters apart, at the same elevation, ± 1.0 meter, with an approximate overall length of 900 meters ± 50 meters shall be used. Horizontal baseline distances to first order accuracy shall be used. A distance measurement shall then be made to each station and corresponding time noted. A distance measurement is defined as the average series of 10 reading interrupted only by repointing at the end of the first five readings. Accuracy and corresponding time measurements not conforming to 3.12.2 and 3.12.3 shall constitute failure of the test.

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c. A baseline consisting of a master monument and remote monument located 7 kilometers, +200 meters or -0 meters, shall be used. SEDME shall be positioned over the master monument, a series of three distance measurements taken, and corresponding times noted. Accuracy and corresponding time measurements not conforming to 3.12.2 and 3.12.3 shall constitute failure of the test.

4.5.2.3 Performance. SEDME shall be tested for accuracy maintainability prior to and after each mechanical and environmental test specified in table I. Accuracy stability checks shall also be made during tests when specified herein. The baseline used for this inspection may be any of the baselines mentioned herein or may be a reference baseline established at the test site. The test shall consist of alining SEDME over the monuments, taking a distance measurement, and noting the time. Accuracy and corresponding time measurements not conforming to 3.12.2 and 3.12.3 shall constitute failure of this test.

4.5.2.4 Stability. SEDME shall be tested for stability by alining over one of the baselines; then with a fully charged internal battery take readings continuously until the battery is expended or the low voltage cut-off is reached as noted on the battery strength meter Accuracy measurements not conforming to 3.12.2 shall constitute failure of this test.

4.5.2.5 Power consumption. While SEDME is being tested in 4.5.2.4, power shall be continuously monitored. Power consumption in excess of that specified in 3.11.4 shall constitute failure of this test.

4.5.2.6 Battery life.

4.5.2.6.1 Main power source. SEDME shall be set up on a baseline used in 4.5.2.2 and measure the distance using the main power source (NiCad battery), until the power is insufficient for operation of the SEDME or the accuracy of the measurements are not in compliance with 3.12.2. The total operating time shall be determined and nonconformance to 3.11.4.1 shall constitute failure of this test.

4.5.2.6.2 External power source. SEDME shall be set up on a baseline used in 4.5.2.2 and measure the distance using the external power source (lead acid battery). Nonconformance to 3.11.4.2 shall constitute failure of this test.

4.5.2.7 High temperature and low temperature operation.

4.5.2.7.1 Test chamber. The high and low temperature operation tests shall be conducted in a chamber of a size that will house SEDME, mounted on its tripod, and accessories and shall permit the inspector to walk around the periphery of the tripod legs in the course of the examination and tests specified herein. Temperature control tolerance for the test chamber shall be ±5 °F or as noted, for all tests specified herein. All temperature gradients shall be 25 ±5 °F/hr.

4.5.2.7.2 High temperature operation. SEDME, mounted in the operating position, with cases, battery, and all accessories shall be placed in a temperature-controlled environmental test chamber. The temperature in the chamber shall be gradually raised at 25 to 120, +5 -0 °F/hr, and maintained at that temperature for not less than 4 hours. Readings shall then be taken according to

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4.5.2.3. Repeat process for 4 cycles. Evidence of damage to any SEDME hardware, failure of SEDME componentry to operate, or nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.7.3 High temperature storage. Upon completion of high temperature operational testing, the SEDME shall be subjected to not less than 160 °F and held at that level for not less than 4 hours. At the end of the 4-hour period, the temperature shall be lowered to ambient and SEDME shall be subjected to the performance test specified in 4.5.2.3 and all accessories shall be examined for damage. Evidence of damage, inability of any accessory to function properly, or nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2 7.4 Solar radiation. SEDME, mounted in the operating position, shall be stabilized in the test chamber at 110 °F, and a source of directed radiant energy simulating solar radiation of not less than 355 Btu per square foot per hour shall be applied to SEDME for a period of not less than 4 hours. At the end of this period, SEDME shall be subjected to the performance test specified in 4.5.2.3. Evidence of damage, inability of any accessory to function, or nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.7.5 Low-temperature operation. SEDME, mounted in the operating position, with cases, battery, and all accessories shall be placed in a temperature-controlled environmental test chamber. The temperature in the chamber shall be gradually lowered 25 to -5, +0 -5 °F/hr, and held at that level for not less than 4 hours. SEDME shall then be subjected to the performance test specified in 4.5.2.3 and inspected for damage (see 6.5). When applicable, the winterization kit shall then be installed on SEDME, batteries removed from the test chamber and the chamber temperature lowered to -25, +0 -5 °F, and held for not less than 4 hours. The winterization kit shall be energized. After a 25-minute warming cycle, the SEDME battery shall be attached; then SEDME shall be inspected and tested as specified in 4.5.2.3. The SEDME battery shall again be removed and the chamber temperature lowered to -45 °F and held for not less than 4 hours. The winterization kit shall be energized. After a warming period of not more than 45 minutes, the SEDME battery shall be attached; then SEDME shall be inspected and tested as specified in 4.5.2.3. Evidence of damage (see 6.5) or nonconformance to 3.4.1 shall constitute failure of these test.

4.5.2.7.6 Low-temperature storage. Upon completion of low-temperature operational testing, the chamber temperature shall be lowered to -50 °F and held for a period of not less than 6 hours. At the end of the 6-hour storage period, the chamber temperature shall be returned to laboratory conditions, allowed to stabilize and dry out for a period of not less than 24 hours at a temperature not less than 85 °F. SEDME shall then be inspected for damage (see 6.5) and tested as specified in 4.5.2.3. Evidence of damage (see 6.5) or nonconformance to 3.4.2 shall constitute failure of the test.

4.5.2.8 Humidity. Place the SEDME, with power off, mounted in the operating position, with its battery, open cases and all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2. The chamber temperature shall be raised to 120, ±2 °F, and 95 percent ±2 percent humidity applied. These conditions shall continue for not less than 24 hours at which time the temperature and humidity shall be removed and conditions allowed to stabilize for 4 hours at ambient conditions. SEDME shall be inspected and tested as specified in 4.5.2.3.

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Evidence of damage (see 6.5) or nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.9 Rainfall. SEDME, mounted in the operating position, with its battery and with cases closed and containing accessories, shall be subjected to rain in accordance with MIL-STD-810, method 506.2, and the following procedure:

The wind source shall be installed and adjusted to produce a horizontal velocity of not less than 30 miles per hour. The test item shall be exposed to simulated wind-driven rainfall at a rate of 3 ±1 inches per hour for not less than 5 minutes. At the end of the 5-minute exposure, install the rain covers, as applicable. The wind shall be maintained at this velocity for not less than 55 minutes. Three of the four sides of the instrument and cases shall be subjected to not less than 15 minutes of wind-driven rain, except the first side, which shall be exposed for not less than 10 minutes. The total test duration, then, is not less than 1 hour. At the end of 1 hour, the rain and wind shall be removed. SEDME instrument, battery, and cases be removed and the test item inspected, with particular attention to evidence of water penetration, such as free water, swelling of material or other deterioration in the instrument of cases, and tested as specified in 4.5.2.3. Evidence of damage (see 6.5), water penetration or nonconformance to 3.4.5 shall constitute failure of the test. Water found inside SEDME which is deemed to have entered during post rain inspection or during removal of covers shall not be considered.

4.5.2.10 Shock.

4.5.2.10.1 SEDME in transport case. SEDME, stowed in transport cases, shall be shock tested in accordance with MIL-STD-819, method 516.3, procedure II, except that the drop height shall be not less than 36 inches. There shall be one flat drop on each of the six faces, and four corner drops. Corners selected for shock shall be optional; however, two corner drops shall be sustained by the bottom shell of the case and the other two sustained by the top shell of the case. In all drops, the case shall be restrained from the secondary shock by bouncing or tumbling. At the conclusion of this shock test, SEDME shall be inspected for damage and tested as specified in 4.5.2.3. Evidence of damage (see 6.5) to SEDME componentry, excluding cases or nonconformance to 3.4.7 shall constitute failure of the test. SEDME cases shall be examined. Inability of the case(s) to protect SEDME componentry shall constitute failure of the test. Damage to cases shall be allowed provided that it does not prohibit the case from continuing its mission. Examples of allowable damage are as follows:

a. Superficial hairline cracks.
b. Deformation of sealing edges less than 1/8 inch end-to-end of case which does not inhibit action, leveling action, or sealing.
c. Wrinkling of corners which will rebound.
d. Locking latches distorted but not loose at mountings or allowing case seal to deteriorate and still able to function.
e. Handles abraded but not deformed.
f. Case abrasion but no through-the-wall rupture.

4.5.2.10.2 SEDME out of transport case. SEDME electronics shall be flat dropped from a height of not less than 4 inches onto the same surface as specified in 4.5.2.10.1 for a total of six "face drops". Each drop shall be initiated by a

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"quick-release" mechanism. In all drops, the electronics package shall be restrained from secondary shock by bouncing or tumbling. Evidence of damage (see 6.5) or nonconformance to 3.16.1 shall constitute failure of this test.

4.5.2.11 Vibration.

4.5.2.11.1 SEDME out of transport case. This vibration test shall be performed with the SEDME electronics package rigidly mounted on the vibration table. If the electronics package case or yoke prohibits direct mounting, then they shall be removed, and the electronics package frame shall be mounted to the vibration table. A mounting fixture may be necessary. The test shall be conducted in accordance with MIL-STD-810, method 514.3 with the following exceptions:

a. Along each of the three major axis, fifteen minutes of cycling (fifteen 1-minute sinusoidal cycles) from 10 to 33 Hz and return, with a 0.10-inch double amplitude or maximum acceleration of 1 g, shall be performed and the major resonance noted.
b. The instrument shall be vibrated 3 minutes at 0.10-inch double amplitude, or 1 g acceleration, on the major resonant points.
c. If there ar no major resonant points below 55 Hz, the instrument shall be vibrated at 55 Hz at a double amplitude of 0.10 inch, or max acceleration of 1 g, for 3 minutes. The accuracy stability test specified in 4.5.2.3 shall be performed immediately preceding and immediately after conclusion of this test. Nonconformance to 3 4.6 or evidence of damage during or as a result of the vibration tests shall constitute failure of this test.

4.5.2.11.2 SEDME in transport case. The vibration test for the instrument and accessories in the transport case(s) shall be performed with the case(s) rigidly mounted to the vibration table. The test shall be conducted n accordance with MIL-STD-810, method 514.3 with the following exceptions:

a. One 5-minute logarithmic cycle for each of the three major axis, from 7 to 200 Hz and return, with a maximum acceleration of 1.5 g, shall be performed.
b. The transport case(s) shall be removed form the vibration table and the test specified in 4.5.2.3 shall be performed Nonconformance to 3 4.6 or evidence of damage shall constitute failure of this test.

4.5.2.12 Magnetic environment. SEDME electronics shall be positioned over a point from which a baseline distance has been established. A magnetic source shall be positioned to produce a magnetic field of not less than 3 gauss (G) at the instrument. Ten distance readings shall be made with the lines of magnetic force directed in each of the following directions: North-south, east-west, and vertical. Readings and corresponding times not conforming to 3.12.1 and 3.12.3 shall constitute failure of the test.

4.5.2.13 Electromagnetic interference. The instruments shall be placed, on a stand 30 inches ±2 inches above the facility floor. No external ground shall be attached. The instrument external power source with external power cable (battery pack) shall be positioned on the stand beside the instrument during both the radiated and conducted tests. Antenna equipment shall be oriented at 1 meter test distance from the instrument with all emissions being measured from one position. The current probe (detecting device) shall be attached to the input power leads to the instrument to measure the conductive emissions. In the event that the input

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power leads are separated, the conductive emissions shall be measured on each lead individually. If the leads are bonded together as a single power cable, both leads shall be measured together. Instruments having interconnect leads other than power source leads shall also be measured for conductive emissions. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.14 Human engineering SEDME shall be tested for human engineering factors specified in 3.7. During the course of all testing herein, test directors shall scrutinize SEDME to determine hardware compatibility to required operator functions. The following functions shall be noted specifically:

a. Suitability for backpack.
b. Operation using arctic mittens.
c. Ease of sighting, tangent running, and clamping.
d. Adaptability to night operation.
e. Location and configuration of all controls.
f. Compatibility of operation with other surveying hardware.
g. Adaptability to charging battery.
h. Interchangeability of SEDME componentry with other hardware during operation.
i. Adaptability to easy "setup" in field operation.
j. Hardware presents no safety hazard from electronics, optics, or mechanical hazard to operator, maintenance personnel, and others.
k. Operation using chemical/biological protective clothing.

4.5.2.15 Reliability. SEDME equipment shall be tested for reliability as specified in 3.5. The reliability test mission shall consist of 10 cycles of the following profile for a total of 200 missions:

a. Start with SEDME packed for transport.
b. Unpack and set up for operation over a known baseline.
c. Perform 10 readings.
d. Dismantle SEDME and repack for transport.
e. Perform manual transport function for not less than 5 feet and return to original position.

NOTE: The use of a pedestal in lieu of a tripod on the baseline is allowed to minimize operator errors in plumbing.

Power for the reliability test shall be either internal volts dc or external power from a 12 volt military battery. The use of two batteries to complete the reliability test is allowable. For the purpose of this test, a mission reliability failure shall be defined as any malfunction, degradation of performance outside specified limits, or personnel and equipment safety hazard which requires more than 30 minutes total operator maintenance time (active and waiting clock time) to correct in order to continue with the defined mission. Repetitive malfunctions occurring more than two times, though corrective within 30 minutes, shall be considered a mission reliability failure. Once the mission is interrupted for more than 30 minutes, it is stopped and counted as a failure. Malfunctions attributable to operator error shall not be considered as chargeable reliability failures. COntinue testing until either and "accept" or "reject" decision is reached. The accept/reject decision for the SEDME mission reliability test shall be based on 200 missions with three failures being cause for rejection of the instruments. For the electronics cycling test portion two instruments

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shall be placed in an environmental chamber in the operational mode. The chamber temperature shall be set and remain at 55 °F for the duration of this test. Power to the instruments shall be supplied by an external source. Measurements shall be initiated and continued for the duration of this test with the following exceptions:

a. The instruments shall be cycled through "self-check" twice each minute.
b. The instruments shall be turned off (using the instrument switch) for a period of 30 ±5 seconds once in each ten-minute period. This procedure shall continue 24 hours each day for an uninterrupted seven-day period. Automatic cycles are allowable for this test. At the end of the seven-day period the instrument shall be removed from the chamber and tested in accordance with 4.5.2.3 Only active operating time shall count toward the seven day test period.

Inability of an instrument to perform any of the operational functions either in the chamber or during operational testing shall constitute failure of this test. Two failures within the seven day time span of this test shall constitute failure of this test. Two successive failures of the Electronic Cycling Sequence shall constitute failure of this test.

4.5.2.16 Maintainability.

4.5.2.16.1 Maintenance support evaluation. Maintenance operation required during first article testing shall be accomplished to determine conformance to 3.8. Failure of the support elements to maintain the SEDME shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance support.

4.5.2.16.2 Maintenance ratio. The maintenance ratio shall be computed during testing specified in 4.5.2.15. Operating time to perform each mission and the active maintenance man-hours required for both scheduled and unscheduled maintenance shall be recorded. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.17 Sand and dust. The SEDME, not in its transport case, without its lens cover, and the closed transport case shall be placed in a sand and dust testing chamber not less than 36 inches high, 36 inches wide, and 36 inches deep. Sand and dust of 140 mesh silica flour shall pass through the chamber at a velocity of 100 to 500 feet per minute. The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at 77 ±5 °F. The relative humidity shall not exceed 30 percent. The SEDME shall be subjected to this test for a period of 15 minutes. Nonconformance to 3.4.9 shall constitute failure of this test.

4.5.2.18 Secure lighting. The SEDME-MR will be tested to ensure that all external lighting conforms to 3.11.7.3.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

132. Materials and containers not as specified for level A or B (see 5.1 and 5.2). Each incorrect material or container shall be considered one defect.
133. Desiccant not placed in the case as specified for level A and B (see 5.1.1).
134. Quantity of desiccant not in accordance with the referenced table for level A and B (see 5.1.1).
135. Wood block not secured in end of box containing the tripod as specified for level A or B (see 5.1.1).
136. Fiberboard boxes not closed and secured in accordance with the referenced methods as specified for level A (see 5.1.1).
137. Marking missing, illegible, incorrect, or incomplete (see 5.3).
138. Strapping not zinc coated for level A.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. All components for which cases are provided shall be placed in their respective cases and the case lids closed and secured. In addition, the case containing the SEDME shall have desiccant placed within the case. The desiccant shall conform to MIL-D-3464 and shall be in the quantity required in accordance with the formula specified in MIL-P-116. The components shall be placed in the transport case as provided for and the case lid closed and secured. Components such as tripods, for which cases are not provided, shall be placed in close-fitting boxes conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade W5c. The box containing the tripod shall have a fitted block of wood secured in the end of the box to prevent the tripod legs from puncturing the box. Each box shall be closed and secured in accordance with method V of the appendix to the box specification.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except that boxes for components shall be type CF, class domestic, variety SE, grade 175. Box closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Level C preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The components comprising a complete SEDME, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification. Unless otherwise specified (see 6.2), strapping shall be class 1, type I or IV, finish B.

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5.2.2 Level B. The components comprising a complete SEDME, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type and strapping need not be zinc coated.

5.2.3 Level C. Level C packaging shall be in accordance with ASTM D 3951 with all components of a complete SEDME consolidated into a single shipping container.

5.3 Marking.

5.3.1 Level A and B. Marking for level A and B shall be in accordance with MIL-STD-129.

5.3.2 Level C. Marking for level C packaging shall comply with requirements of ASTM D 3951. In addition, weight and cube data shall be marked on each shipping container.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The SEDME shall be used to determine slope distances for field artillery tactical surveys, and for engineer topographic control and construction surveys.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required for inspection and approval (see 3.2).
d. Color when other than specified (see 3.10).
e. When a winterization kit, thermometer, altimeter-barometer and meteorological correction data are required (see 3.1, 3.17 and 3.18).
f. When tripods are required (see 3.1.1 and 3.14).
g. When a backpack is required (see 3.15).
h. Level of preservation and level of packaging required (see 5.1 and 5.2).
i. When strapping other than finish B is required (see 5.2.1).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of two or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, early delivery of SEDME systems for training development, and contractor's maintenance and operational manual to be furnished with each instrument.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of instruments.

6.6 Definition.

6 6.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the SEDME with provisions of this specification.

6.6.2 Maintenance ratio. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time.

6.7 Safety. The contracting officer should arrange to furnish to the contract a copy of TB MED 524 and should require the contractor to furnish certified radiometric data of the infrared radiation output of a representative surveying instrument of the contractor's own manufacture that complies with this specification to the contracting officer of technical representative in accordance with AR 40-5 for the purpose of determining that all applicable safety requirements specified in 3.6 have been complied with.

6.8 Subject term (key word) listing.

distance measuring equipment
electro-optical package
amplitude modulated light beam
slope distance determination
field artillery tactical survey
construction survey

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army- ME

Preparing activity:
Army - ME

Project 6675-A824

Review activity:
DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-S-0053046A(ME)	2. DOCUMENT DATE (YYMMDD) 940422

3. DOCUMENT TITLE
SURVEYING EQUIPMENT, ELECTRIC DISTANCE MEASURING, MEDIUM-RANG, (SEDME-MR)

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION	
c ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340	

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